Functional Equations — problem 11.26

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Find all continuous functions $f : \mathbb{R} \mapsto \mathbb{R}_+$ which transform three terms of the arithmetic progression $x, x+y, x+2y$ into corresponding terms $f(x), f(x+y), f(x+2y)$ of a geometric progression, that is,

$$[f(x+y)]^2 = f(x) \cdot f(x+2y).$$